

Step 1: Define the Problem: (What, When, Where, First time observed, Total population)

Machine spare / dispatch history is not easily accessible to Maintenance Techs.

Step 2: Problem Breakdown (Pictures, Drawings, Charts, etc. to illustrate problem as much as possible)

- The L2L information is available to the maintenance technicians but not readily accessible from the breakdown location.

- Without the available information, waiting and downtimes are longer than necessary.

- Operators and Maintenance are performing their tasks with unnecessary set-backs.

Current:

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graph LR; A((Operator calls Supervisor)) --> B((Supervisor calls Maintenance)); B --> C((Maintenance Dispatch)); C --> D((Maintenance arrives at Breakdown)); D --> E((Maintenance Assesses Issues)); E --> F{Dispatch Place?}; F -- No --> G((Maintenance Checks Dispatch for Parts)); G --> F; F -- Yes --> H{Part Correct?}; H -- No --> I((Maintenance Returns to get Parts)); I --> J((Maintenance Returns to Breakdown)); H -- Yes --> K((Maintenance Repairs)); K --> L((Maintenance Closes Dispatch));
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Intermediate:

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graph LR; A((Operator calls Supervisor)) --> B((Supervisor Dispatch)); B --> C((Maintenance Checks Dispatch for Recommended Parts)); C --> D((Maintenance Dispatch)); D --> E((Maintenance arrives at Breakdown)); E --> F((Maintenance Assesses Issues)); F --> G{Maintenance Validates Parts}; G -- No --> H((Maintenance Returns to get Parts)); H --> I((Maintenance Returns to Breakdown)); G -- Yes --> J((Maintenance Repairs)); J --> K((Maintenance Closes Dispatch));
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Ideal:

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graph LR; A((Operator Dispatch)) --> B((Maintenance Checks Dispatch for Recommended Parts)); B --> C((Maintenance Dispatch)); C --> D((Maintenance arrives at Breakdown)); D --> E((Maintenance Assesses Issues)); E --> F{Maintenance Validates Parts}; F -- No --> G((Maintenance Returns to get Parts)); G --> H((Maintenance Returns to Breakdown)); F -- Yes --> I((Maintenance Repairs)); I --> J((Maintenance Closes Dispatch));
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[Process Maps](#)

Decrease machine breakage time and report machine failure more quickly and effectively.

Step 4: Root Cause Analysis (Identify possible root causes and validate their contribution to the problem)

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graph LR; Machine[Machine] --> Main[Inefficiency in Maintenance and PM]; Method[Method] --> Main; Measurement[Measurement] --> Main; Method --> Inefficiency[Inefficiency in Identifying Parts]; Method --> NoDocs[No Supporting Documents for Machine History]; Method --> Response[Equipment Failure Response Time];
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Step 5: Develop and Prioritize Countermeasures

Prioritize implementation of countermeasures by using the calculated value in Overall column = Cost score X Feasibility score X Effectiveness score. Prioritize Higher Overall First								
Score 3 = Low Cost, Easy to Implement, Effective Score 2 = Mid \$\$, long time to Implement, Some what Effective Score 1 = High \$\$\$, Hard to Implement, Low Effectiveness								
Cause No.	Root Cause	Countermeasure	Cost \$	X	Feasibility X	Effectiveness	= Overall	Comments
1	Equipment Failure Response Time	Improved dispatch reporting with QR codes	3		3	2	18	
2	Inefficiency in Identifying Parts	Have part/spare information readily available at breakdown location	3		3	2	18	
3	No Supporting Documents for Machine History	Include all necessary documents and information in L2L and accessible from the breakdown location	3		3	2	18	

Step 6: Implement Countermeasures

ID	Countermeasure Implementation Task	Start	Finish	Responsible	Weeks after start date (Check the boxes for duration to complete task)											
					1	2	3	4	5	6	7	8	9	10	11	12
1	Create QR Codes for L2L Quick Links	6/20/2025	6/27/2025	Michael H	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
2	Put QR Codes on Assembly Lines	7/2/2025	7/22/2025	Michael H	☒	☒	☒	☒	☐	☐	☐	☐	☐	☐	☐	☐
3	Take Pictures of Equipment to Update Links	7/2/2025	7/2/2025	Austin L	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
4					☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
5					☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
6					☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Step 7: Countermeasure Effectiveness (Validate using data)

Set the Goal+limits	Weeks after start date (Trend the performance every week)											
	1	2	3	4	5	6	7	8	9	10	11	12
	Actual:											
- Validate Countermeasure(s) had the expected impact on the problem defined in <u>Step 1</u>. - Trend the actual levels weekly and compare to the Goal/Target defined in <u>Step 3</u>.												

Step 8a: Standardize Successes (If a question answer is "No", the appropriate task needs to be completed to answer "Yes")

No	Process Changes	Yes	No	N/A	No	Tools / Equipment Change	Yes	No	N/A
1	Changes are reflected in the Work Instructions	☐	☐	☒	1	Drawings for Tools / Equipment reflect the changes	☐	☐	☒
2	The Issue will not occur when following the work instruction	☐	☐	☒	2	Operators trained on the use of new tools/equipment	☒	☐	☐
3	The operator can identify/detect the issue	☒	☐	☐	3	Operators trained on Autonomous Maintenance	☒	☐	☐
4	Changes are reflected in the control plan	☐	☐	☒	4	Changes reflected on the Preventive Maintenance	☒	☐	☐
5	Changes are reflected in the PFMEA	☐	☐	☒	Other Forms and Next Steps:				
6	Operators/Leads/Supervisors trained in new process	☒	☐	☐					

Step 8b: Lessons Learned and Shareability (Actual Outcome)

Implementation Contact and Information		Can this countermeasures be implemented in similar processes in the plant or in other plants? Which ones?	Yes
Plant Contact:		Lessons learned / Outcome:	
Phone number:		The biggest takeaways from this project was the importance of escalation and making changes in order to reduce downtime. There were a great deal of issues with escalation and it was very apparent a change needed to be made. This countermeasure of using QR codes to make dispatches quicker and more fluid will reduce downtime significantly.	
Cost per line/area:			
Vendor / Supplier:			
Drawing / ECO:			

AA011-Da, Rev (TBD)

Date: DD MMM YYYY; ECO #XXXXXXX; C. Fretwell /